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Inventor(s)	Paukamainen; Hannu et al.

QUICK-COUPLING HOOK WITH FACILITATED OPERATION

Abstract

A quick-coupling hook having a hook body, a securing lock, an actuating component coupled with the securing lock for common movement, a pre-tensioning spring, a first hook body-side abutment formation on a first side of the actuating component, and a second hook body-side abutment formation on a second side of the actuating component opposite to the first side, where the pre-tensioning spring extends between a hook body-side and an actuating component-side spring bearing and pre-tensions the actuating component towards its operating position, where the actuating component in its operating position abuts against the first and against the second hook body-side abutment formation, where it is provided that in a reference state of the quick-coupling hook, in which the actuating component is in an operating position and the securing lock in a decoupling-preventing securing position, a virtual straight connecting line passing through both the hook body-side spring bearing and the actuating component-side spring bearing of the pre-tensioning spring exhibits at least double the distance from the first hook body-side abutment formation as it does from the second hook body-side abutment formation.

Inventors: **Paukamainen; Hannu (Tikkakoski, FI), Mäkelä; Lasse (Jyväskylä, FI)**

Applicant: **JOST-Werke Deutschland GmbH** (Neu-Isenburg, DE)

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Background/Summary

[0001] This Application claims priority in German Patent Application DE 10 2024 106 098.0 filed on Mar. 2, 2024, which is incorporated by reference herein.
[0002] The present invention concerns a quick-coupling hook, in particular for agricultural vehicles. Such a quick-coupling hook, which is used for example as part of a liftable and lowerable tractor-side three-point coupling for coupling agricultural working tools, such as for instance ploughs, harrows, and the like, comprises: [0003] i. A hook body with a hook mouth, where the hook mouth exhibits an accommodating region accessible through an opening region, [0004] ii. A securing lock which is accommodated at the hook body and which is displaceable relative to the hook body between a securing position and a releasing position, where in the securing position the securing lock projects further into the opening region than in the releasing position, [0005] iii. An actuating component which is accommodated at the hook body and which is moveable relative to the hook body between an operating position and a setting-up position, where the securing lock and the actuating component are coupled with one another for common movement in such a way that when the actuating component is in its setting-up position, the securing lock is in its releasing position, and that when the actuating component is in its operating position, the securing lock is in its securing position, [0006] iv. A pre-tensioning spring which extends between a hook body-side spring bearing and an actuating component-side spring bearing and which pre-tensions the actuating component towards its operating position, [0007] v. A first hook body-side abutment formation on a first side of the actuating component, and [0008] vi. A second hook body-side abutment formation on a second side of the actuating component which is opposite to the first side.
[0009] The actuating component abuts in its operating position against the first and against the second hook body-side abutment formation. The actuating component exhibits a force application formation which is configured, through force application thereon, to move the actuating component starting from the operating position against the effect of the pre-tensioning spring in the direction of the setting-up position.
[0010] The quick-coupling hook is described in the present application, unless stated otherwise, in a reference state in which the actuating component is in its operating position free from external force application on the quick-coupling hook. Therefore at least in the reference state the aforementioned first and second hook body-side abutment formations are situated on the first side and on the second side opposite to the first of the actuating component.
[0011] The description of the quick-coupling hook in the reference state does not mean that the technical features specified in the description apply only to the reference state. They apply, however, at least in the reference state of the quick-coupling hook.
BACKGROUND OF THE INVENTION
[0012] Such a quick-coupling hook of the type mentioned in the beginning is known from EP 1 849 632 A1. As is usual for quick-coupling hook, the known quick-coupling hook should allow it to establish in the shortest possible time a temporary positive-locking connection with a counter-coupling component, often a counter-coupling ball, to maintain it securely, and to disengage it again in the shortest possible time.
[0013] At the known quick-coupling hook, the actuating component can be secured in its setting-up position through an abutment engagement of a latching counter-formation of the actuating component with a hook body-side latching formation. In the setting-up position, a counter-coupling

component previously coupled with the quick-coupling hook can be removed out of the accommodating region through the opening region of the hook mouth. Consequently, equipping or re-equipping of the vehicle carrying the quick-coupling hook can take place in the setting-up position of the actuating component.

[0014] In the operating position of the actuating component, a counter-coupling component accommodated in the accommodating region is secured through the securing lock, which is then in the securing position, against leaving the accommodating region and thereby against disengaging of the coupling connection. A vehicle carrying the quick-coupling hook can then operate with a coupled working tool.

[0015] A drawback in the known quick-coupling hook is the complicated movement sequence which an operator has to perform manually through finger engagement at the force application formation configured as an annular eyelet in order to move the actuating component securely from its operating position into its setting-up position and there secure it at the designated hook body-side latching formation.

[0016] To this end the operator has to pull out the actuating component accommodated in the operating position in the hook body translationally from the hook body and has to tilt the actuating component in the pulled-out state at the correct point in time against the pre-tensioning effect of the pre-tensioning spring. Untrained operators can require several attempts for this until they successfully bring the actuating component into the setting-up position, which thwarts the object of a quick-coupling hook.

SUMMARY OF THE INVENTION

[0017] It is, therefore, the task of the present invention to facilitate the operating of the quick-coupling hook for moving the actuating component from the operating position into the setting-up position.

[0018] The present invention solves this task in the quick-coupling hook mentioned at the beginning thereby that in the mentioned reference state a virtual straight connecting line passing through both the hook body-side spring bearing and the actuating component-side spring bearing exhibits at least double the distance from the first hook body-side abutment formation as it does from the second hook body-side abutment formation.

[0019] The first and the second hook body-side abutment formation and the linking of the actuating component to the pre-tensioning spring at the actuating component-side spring bearing define position and orientation of the actuating component in its operating position. Starting from the operating position, the actuating component can be moved only translationally in the direction towards the setting-up position in a direction running between the first and the second hook body-side abutment formation.

[0020] In the quick-coupling hook discussed here, just as in the quick-coupling hook known from EP 1 849 632 A1, the actuating component is accommodated in a recess of the hook body, where in the reference state the actuating component is situated deeper in the recess than in its setting-up position.

[0021] The virtual straight connecting line which passes through the hook body-side spring bearing and through the actuating component-side spring bearing, indicates the line of action of the force with which the pre-tensioning force exerted by the pre-tensioning spring acts on the actuating component. As a result of the distance of the virtual straight connecting line from the first hook body-side abutment formation being at least double the distance of the virtual straight connecting line from the second hook body-side abutment formation, there is generated when removing the actuating component from its operating position through the pre-tensioning spring not only a pre-tensioning force, but a pre-tensioning force acting eccentrically on the actuating component. In addition to the pre-tensioning effect, this eccentric pre-tensioning force generates a torque acting on the actuating component, which as a tilt moment effects a tilt of the actuating component about a tilt axis running transversely, preferably orthogonally, to the trajectory of the translational movement of the actuating component. For the operator, therefore, it suffices to remove the actuating component from the operating position against the pre-tensioning effect of the pre-tensioning spring. A tilting movement which is desirable in particular for automatic fixing of the actuating component in its setting-up position, sets in by itself due to the arrangement of the two spring bearings of the pre-tensioning spring. Even an operator who thus far has not encountered the quick-coupling hook and therefore is unfamiliar with it, is therefore with the greatest probability already successful in the first attempt to move the actuating component from the operating position into the setting-up position and fix it there.

[0022] The tilting effect generated by the pre-tensioning spring at the actuating component is greater, the greater the distance of the virtual connecting axis from the first hook body-side abutment formation and the smaller the distance of the virtual connecting axis from the second hook body-side abutment formation. Therefore, the distance of the virtual connecting axis from the first hook body-side abutment formation is preferably at least five times greater than the distance of the virtual connecting axis from the second hook body-side abutment formation, especially preferably at least 10 times greater, and even more preferably at least 15 times greater.

[0023] To prevent unnecessarily high friction between the actuating component and the hook body-side abutment formations, which could hamper a movement of the actuating component out of its operating position, it is preferably provided that the virtual straight connecting line which is conceived to be running out via the two spring bearings, runs in the reference state of the quick-coupling hook between the first and the second hook body-side abutment formation. The virtual straight connecting line can be a tangent to the second hook body-side abutment formation, such that in this case the distance between the virtual straight connecting line and the second hook body-side abutment formation is zero. In this case, the distance of the virtual straight connecting line from the first hook body-side abutment formation is deemed to be greater by an infinitely large factor than the non-existing distance of the virtual straight connecting line from the second hook body-side abutment formation.

[0024] The virtual straight connecting line can also run beyond the second hook body-side abutment formation, such that both the first and the second hook body-side abutment formation lie on the same side of the virtual straight connecting line. In this case, the magnitude of the distance of the virtual straight connecting line from the second hook body-side abutment formation is used. A negative distance should not be used for analyzing the ratio of the two distances.

[0025] In order to fix the actuating component in its setting-up position, in the quick-coupling hook being discussed here too, it is preferably provided that the quick-coupling hook exhibits a hook body-side latching formation with which an actuating component-side latching counter-formation of the actuating component, in a setting-up state of the quick-coupling hook in which the actuating component is in the setting-up position, is engaged in order to hold the actuating component in the setting-up position against the effect of the pre-tensioning spring. Since a counter-coupling component accommodated in the hook mouth in the reference state of the quick-coupling hook is secured in the reference state against removal from the hook mouth and the quick-coupling hook is thus operationally ready, the reference state is also an operational state of the quick-coupling hook.

[0026] The actuating component is preferably a flat component which in two spatial directions orthogonal to one another exhibits considerably greater dimensions, approximately greater by at least a factor of three to four, than in its thickness direction. The contour of the actuating component can be complex, since the actuating component unifies in itself several functionalities and at the same time should exhibit the lowest weight possible. Thus preferably the actuating component-side latching counter-formation is configured as a contour section of the contour of the actuating component and therefore as part of its lateral surface. The force application formation of the actuating component can likewise impact the contour of the actuating component and at least section-wise be part of the contour.

[0027] As a flat component, the actuating component is preferably arranged in such a manner at and in the hook body that its thickness direction is oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth. Preferably, the actuating component executes between its operating position and its setting-up position a planar movement in the sense that the respective orientations of its thickness direction both in the operating position and in the setting-up position, preferably also in all intermediate positions, are parallel to one another.

[0028] In the reference state of the quick-coupling hook, the latching formation preferably lies on the same side of the actuating component as the first hook body-side abutment formation. The actuating component can then in the operational state abut against the first hook body-side abutment formation, and in the setting-up state can on the same side be engaged with its latching counter-formation with the hook body-side latching formation.

[0029] To avoid an unnecessarily large number of components for forming the quick-coupling hook, the hook body-side latching formation is preferably also the first hook body-side abutment formation. In a preferred development of the present invention, the hook body-side latching formation and thereby especially preferably also the first hook body-side abutment formation is a pin, for example a cylindrical or polyhedral pin, which especially preferably passed through a recess in the hook body in which the actuating component at least in its setting-up position is accommodated. The pin configured as a latching formation preferably runs parallel to the thickness direction of the actuating component at the hook body. Further preferably, the latching formation is a rotation-symmetrical component with respect to an axis of rotational symmetry parallel to the thickness direction of the actuating component, such as for instance a cylindrical pin, such that its orientation does not matter during its assembly.

[0030] Alternatively, the hook body-side latching formation and/or the first hook body-side abutment formation can be configured integrally with the hook body as a body formation of the latter.

[0031] The engagement of the actuating component-side latching counter-formation with the hook body-side latching formation is preferably a simple to establish abutment engagement, in which preferably an exterior surface section of a lateral surface of the actuating component as the actuating component-side latching counter-formation abuts against a lateral surface of the latching formation. For a stable abutment engagement, the latching counter-formation can be a section-wise negative reflection of the latching formation. For example, when the latching formation is formed by a cylindrical pin, the latching counter-formation can exhibit a negative part-cylindrical surface which in engagement with the latching formation surrounds the latter along a wrap angle of at least approximately 45°, preferably at least approximately 60°, especially preferably at least approximately 80°. In this case, the actuating component can be pivoted around the latching formation with the latching formation as a pivot axis, without having to disengage the abutment engagement of the latching counter-formation with the latching formation or actually undesirably disengaging it.

[0032] When, in the setting-up state of the quick-coupling hook, the actuating component-side latching counter-formation is in abutment engagement with the hook body-side latching formation, then in order to be able to hold the actuating component securely in its setting-up position despite merely an abutment engagement, the actuating component preferably exhibits a setting-up abutment counter-formation lying at a distance from the latching counter-formation which in the setting-up state abuts against a hook body-side setting-up abutment formation. In order to be able to secure the setting-up position of the actuating component which is defined solely through abutment engagements between formations of the actuating component on the one hand and of the hook body on the other, the virtual straight connecting line runs in the setting-up state at a distance from the latching formation in order to initiate a tilt moment in the actuating component which pushes the actuating component about the latching formation into an abutment engagement with the hook body-side setting-up abutment formation without undoing the abutment engagement with the latching formation.

[0033] In the setting-up state, the distance of the virtual straight connecting line from the hook body-side setting-up abutment formation preferably equals at most double the distance of the virtual straight connecting line from the hook body-side latching formation. Especially preferably, in the setting-up state the distance of the virtual straight connecting line from the hook body-side setting-up abutment formation does not exceed 1.25 times the distance of the virtual straight connecting line from the hook body-side latching formation.
[0034] In particular when, but not only when, the abutment engagement of the actuating component-side latching counter-formation with the hook body-side latching formation allows pivoting of the actuating component about the latching formation while maintaining the abutment engagement, then in the setting-up state the distance of the virtual straight connecting line from the hook body-side setting-up abutment formation is preferably smaller than the distance of the virtual straight connecting line from the hook body-side latching formation, since then the distance of the actuating component-side spring bearing from the actuating component-side latching counter-formation or the hook body-side latching formation, as the case may be, establishes a load arm on whose length, for a specified pre-tensioning force of the pre-tensioning spring, the tilt moment acting around the latching formation, initiated by the pre-tensioning spring in the actuating component, depends.
[0035] Thus the actuating component in its setting-up position can also be securely fixed at the hook body when the hook body-side setting-up abutment formation is configured as parallel to the virtual straight connecting line and thereby parallel to the direction of action of the pre-tensioning force of the pre-tensioning spring or only with a small tilt to it, such that it can only in a negligible manner support directly a pre-tensioning force of the pre-tensioning spring. In contrast, an abutment force resulting from the tilt moment initiated about the latching formation can always support the hook body-side setting-up abutment formation regardless of its orientation relative to the pre-tensioning force.
[0036] In a preferred concrete embodiment, in the reference state the first hook body-side abutment formation is situated on the side of the actuating component facing away from the hook mouth and the second hook body-side abutment formation is situated on the side of the actuating component facing towards the hook mouth.
[0037] The pre-tensioning force of the pre-tensioning spring preferably acts as parallel as possible to the direction of movement of the actuating component, such that as large a fraction as possible of the pre-tensioning force does actually act in the desired direction, namely towards the operating position. Further preferably, the actuating component is moved between its operating position and its setting-up position as parallel as possible to a mouth center plane oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth and running centrally both through the opening region and through the accommodating region. The mouth center plane preferably divides the cross-sectional area of the accommodating region surrounded by the hook mouth into two equally large area parts. Experts can readily visualize the mouth center plane of the hook mouth. Along an insertion trajectory lying in the mouth center plane there can moreover a counter-coupling component which is to be coupled with the quick-coupling hook be brought into the accommodating region of the hook mouth. For the aforementioned reasons, therefore, it is preferably provided that in the reference state and/or in the setting-up state, especially preferably also in an arbitrary intermediate position of the actuating component, the virtual straight connecting line encloses an angle not exceeding 15° with the mouth center plane and/or with a virtual insertion trajectory of a counter-coupling component to be coupled with the quick-coupling hook into the accommodating region.
[0038] Likewise, what is important for the ergonomics of an actuation of the actuating component is the arrangement of the quick-coupling hook being discussed here and its orientation in the mounted state on the vehicle carrying it. The hook body preferably exhibits an—ordinarily standardized—connection surface for connecting to a vehicle-side carrier. Especially preferably,

this connection surface is a connection plane or a surface which is uncurved in at least in one direction. Ordinarily, the quick-coupling hook is welded by means of the connection surface onto a carrier of an agricultural vehicle. Since, however, here the quick-coupling hook is being discussed per se and without orientation relative to the vehicle carrying it, the connection surface offers the best possible indication for assessing the movement trajectory of the actuating component later in a state mounted on a vehicle. Preferably, the virtual straight connecting line encloses in the reference state and/or in the setting-up state, especially preferably also in an arbitrary intermediate position of the actuating component, with the connection surface an angle not greater than 10°. Further preferably, the connection surface encloses with the previously discussed mouth center plane and/or with the previously discussed insertion trajectory of a counter-coupling component into the accommodating region of the hook mouth an angle not greater than 10°.

[0039] In order to be able to move the actuating component as securely and disruption-free as possible at least from its operating position into its setting-up position, preferably however also in the opposite direction, according to a development of the present invention the actuating component exhibits a sliding surface which runs from an actuating component-side first abutment counter-formation with which in the reference state the actuating component abuts against the first hook body-side abutment formation up to the actuating component-side latching counter-formation. The actuating component can preferably slide with its sliding surface during its movement from the operating position into the setting-up position along an exterior surface of the latching formation, thus serving an operator manually moving the actuating component together with the latching formation as movement guidance, approximately in the manner of a cam or slotted guide.

[0040] In principle, the securing lock can be coupled arbitrarily kinematically with the actuating component. In a preferred embodiment of the quick-coupling hook being discussed here, the securing lock is simply and securely articulated pivotably about a virtual linkage axis at the actuating component. The linkage axis preferably runs in the thickness direction of the actuating component, thus preferably also in the direction in which the latching formation penetrates through the accommodating space of the hook body in which the actuating component is accommodated at least in its operating position.

[0041] For a stable position and orientation of the actuating component in its setting-up position, in the setting-up state of the quick-coupling hook the virtual linkage axis preferably lies on the same side as the latching formation with respect to a reference plane containing the virtual straight connecting line and oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth. The securing lock can be moveable only with difficulty relative to the hook body just immediately before reaching the setting-up position, due to abutment engagements of the securing lock with the hook body and thereby effected friction. The virtual linkage axis, so to speak spatially fixed through the securing lock, can then be a tilt axis for the actuating component on reaching the abutment engagement of the actuating component-side latching counter-formation with the hook body-side latching formation. The position of latching counter-formation and linkage axis in the setting-up state on the same side of the virtual connecting axis is then advantageous for the movement sequence to reach the mentioned abutment engagement of latching formation and latching counter-formation.

[0042] In principle, the force application formation is preferably provided for a manual actuation and/or movement respectively of the actuating component. To this end, the force application formation can comprise a rear grip formation for a manual finger grip or hand grip. Such a rear grip formation can be a handle or an annular eyelet or a reach-through aperture in the actuating component. In case of an annular eyelet or a reach-through aperture, it can preferably be gripped through in the thickness direction of the actuating component. Alternatively, a rod or a hand grip can serve as manual force engagement which projects out from the actuating component and can be gripped around by an operator.

[0043] Additionally or alternatively, the force application formation can comprise and/or be, as the case may be, a coupling formation of the actuating component for coupling with an output element of an actuator. When the actuating component should be moved through an actuator at least from the operating position into the setting-up position, the force application formation can be an arbitrary coupling formation for coupling the output element of the actuator with the actuating component, such as part of a ball joint, such as for instance a ball socket, or the like.

[0044] An advantageous tilting effect in the interaction of the pre-tensioning spring with an actuating force exerted on the actuating component in the direction away from its operating position can thereby be achieved that in the reference state a large part of the actuating component lies on the same side of the virtual straight connecting line on which the latching formation and/or the first hook body-side abutment formation also lies. Since the actuating component is a three-dimensional physical object, whereas the virtual straight connecting line is a linear shape, the arrangement of a large part of the actuating component on one side of the virtual straight connecting line is more simply expressed through a reference plane containing the virtual straight connecting line, which in contrast to the straight connecting line is capable of virtually intersecting the entire actuating component. Therefore it is preferably provided that in the reference state at least 75% of the actuating component lie on the same side with respect to a reference plane containing the virtual straight connecting line and oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth as the latching formation and/or the first hook body-side abutment formation. In particular, at least 75%, preferably at least 85% of the force application formation of the actuating component lie on this side of the reference plane, in order to effect under a force application to the force application formation a desirable tilt moment at the actuating component.

[0045] As already indicated above, the quick-coupling hook can exhibit an actuator in order to move the actuating component from the operating position into the setting-up position. The actuator can be an electromechanical, electromagnetic, pneumatic, or hydraulic actuator.

[0046] These and other objects, aspects, features and advantages of the invention will become apparent to those skilled in the art upon a reading of the Detailed Description of the invention set forth below taken together with the drawings which will be described in the next section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0047] The invention may take physical form in certain parts and arrangement of parts, a preferred embodiment of which will be described in detail and illustrated in the accompanying drawings which forms a part hereof and wherein:

[0048] FIG. **1**A schematic side view of an embodiment according to the invention of the quick-coupling hook of the present application in the reference state without lateral covering, in order to show the components of the quick-coupling hook accommodated in the hook body,

[0049] FIG. **2**A schematic side view as in FIG. **1**, but with actuating component moved away from the operating position,

[0050] FIG. **3**A schematic side view as in FIGS. **1** and **2** with actuating component moved out even further, where the actuating component is just before reaching its setting-up position, and

[0051] FIG. **4**A schematic side view as in FIGS. **1** to **3** with the actuating component in the setting-up position and the quick-coupling hook in the setting-up state, respectively.

DESCRIPTION OF PREFERRED EMBODIMENTS

[0052] Referring now to the drawings wherein the showings are for the purpose of illustrating preferred and alternative embodiments of the invention only and not for the purpose of limiting the same, in FIG. **1**, an embodiment according to the invention of a quick-coupling hook is labelled generally by **10**. The quick-coupling hook **10** of FIG. **1** is not depicted completely, since a covering as part of the hook body **12** facing towards the observer of FIG. **1** is left out in order to be able to depict the components accommodated in the interior of the hook body **12**: securing lock **14**, actuating component **16**, pre-tensioning spring **18**, latching formation **20**, and hook body-side spring bearing **22**. Of the hook body **12**, therefore, there is depicted in FIGS. **1** to **4** only a part-body **13**, which however forms the largest part of the hook body **12**.

[0053] For better orientation, a Cartesian triad in the top right-hand corner of FIGS. **1** to **4** indicates the vehicle axes of a vehicle not depicted in the drawings which carries the quick-coupling hook **10**. The vehicle axes are the roll axis Ro, the pitch axis Ni, and the yaw axis Gi. The arrow of the roll axis Ro points in the forward travel direction of the vehicle. The quick-coupling hook **10** thus projects in the depicted embodiment example from the rear or from a carrier at the rear, as the case may be, of an agricultural vehicle in the reverse travel direction.

[0054] The hook body **12** or more precisely its part-body **13** depicted in FIGS. **1** to **4** exhibits as a central coupling formation a hook mouth **24**, whose accommodating region **26** is surrounded on three sides by a hook section **28** curved in the shape of a hook. The accommodating region **26** is accessible through an opening region **34** to a counter-coupling component **30** depicted in FIG. **1** only in rough schematic form, such as a spherical section of a counter-coupling formation, along an insertion trajectory **32** determined by the shape of the through hook mouth. The insertion trajectory **32** lies in FIGS. **1** to **4** in a mouth center plane **33** orthogonal to the respective drawing plane of FIGS. **1** to **4** which is oriented orthogonally to a cross-sectional area **26***a* of the accommodating region **26** surrounded by the hook mouth **24** and running centrally both through the opening region **34** and through the accommodating region **26**. The cross-sectional area **26***a* of the accommodating region **26** surrounded on three sides by the hook mouth **24** or more precisely by the hook section **28** is depicted hatched in FIG. **1**.

[0055] A lead-in chamfer **36** at the hook section **28** in the opening region **34** facilitates the insertion of the counter-coupling component **30** into the accommodating region **26**.

[0056] In FIG. **1**, the securing lock **14** is in its securing position in which it protrudes maximally into the opening region **34** and prevents the escape of a counter-coupling component **30** accommodated in the accommodating region **26** along the insertion trajectory **32** but in the opposite direction, more precisely along the mouth center plane **33** through the opening region **34**. Through the protrusion of the securing lock **14** into the opening region **34**, the clear width of the opening region **34** is decreased to the extent that the counter-coupling component **30** does not pass through the remaining gap between the lead-in chamfer **36** and the securing lock **14**. An abutment of the securing lock **14** in its securing position against a securing surface **38***a* of a hook body section **38** situated over the securing lock **14** in its securing position prevents a counter-coupling component **30** accommodated in the accommodating region **26** displacing the securing lock **14** from its securing position.

[0057] In the opposite direction, however, a counter-coupling component **30** is capable of reaching the accommodating region **26** despite a securing lock **14** arranged in its securing position, since the counter-coupling component **30** during opposite movement from outside into the accommodating region **26** can push the securing lock **14** out of its securing position.

[0058] The securing lock **14** is articulated pivotably at the actuating component **16** about a linkage axis A orthogonal to the drawing plane of FIG. **1**.

[0059] Solely for the sake of completeness let it be mentioned that a counter-coupling component **30** moved from outside, i.e. in the case of FIG. **1** from above, along the mouth center plane **33** or along the insertion trajectory **32** respectively into the accommodating region **26** hits a concave contact surface **14***a* of the securing lock **14**. The securing lock **14**, which in its securing position abuts against a cylindrical tip formation **40** in the hook body **12**, can tip about the tip formation **40**, i.e. about a tip axis parallel to the linkage axis A, under the load application by the counter-coupling component **30** which is moved into the hook mouth **24**. Hereby the longitudinal end of the securing lock **14** protruding into the opening region **34** is pressed downwards and the longitudinal end articulated at the actuating component **16** correspondingly displaced upwards. With the thus effected ejection movement of the actuating component **16** out of the recess **42** in the hook body **12** which accommodates it in the operating position shown in FIG. **1**, the securing lock **14** can be pushed by the counter-coupling component **30** into the channel **44** which in FIG. **1** is bounded upwards by the hook body section **38** and downwards by the hook section **28**, whereby the opening region **34** becomes passable for the counter-coupling component **30**.

[0060] The pre-tensioning spring **18**, which in a manner usual per se is a tension spring, is suspended at its hook body-remote end at an actuating component-side spring bearing **46**, for instance a fastening eyelet which passes through the actuating component **16** in the thickness direction, that is, in the depicted embodiment example orthogonally to the drawing plane of FIG. **1**. Through the choice of the location of the hook body-side spring bearing **22** at the hook body **12** on the one hand and of the location of the actuating component-side spring bearing **46** at the actuating component **16** on the other, the pre-tensioning force exerted by the pre-tensioning spring **18** on the actuating component **16** can be determined constructionally in its direction of action. Through the choice of the pre-tensioning spring **18**, the pre-tensioning force is determined magnitude-wise as a function of its excursion.

[0061] A virtual straight connecting line **48** which connects the two spring bearings **22** and **46** with one another and of which only a section is shown in FIG. **1**, indicates the direction of action of the pre-tensioning force V applied by the pre-tensioning spring **18** to the actuating component. The arrow of the pre-tensioning force V is meant to indicate the pre-tensioning force in FIG. **1** merely qualitatively. The virtual straight connecting line **48** runs in the depicted embodiment example in parallel to the drawing plane of FIG. **1** or at least in a plane **50** orthogonal to the drawing plane of FIG. **1**. The cross-sectional area **26***a* of the accommodating region **26** is likewise oriented in parallel to the drawing plane of FIG. **1**. The aforementioned plane **50** is, therefore, likewise orthogonal to it.

[0062] The pre-tensioning spring **18** urges the actuating component **16** in its operating position shown in FIG. **1**. The pre-tensioning spring **18** thereby also pre-tensions the securing lock **14** in its securing position shown in FIG. **1**. FIG. **1** therefore shows the quick-coupling hook **10** in the reference state mentioned in the descriptive introduction.

[0063] In the present case, however, it is less the displacement of the actuating component **16** through the securing lock **14** that is of interest but rather conversely the displacement of the securing lock **14** through a manual actuation of the actuating component **16**.

[0064] At its upper longitudinal end projecting out of the recess **42**, the actuating component **16** exhibits an annular eyelet **52** as a force application formation **54**. The reach-through area **56** surrounded by the annular eyelet **52** is likewise oriented in parallel to the drawing plane of FIG. **1**. The annular eyelet **52** is suitable and intended to be gripped from behind by one or two fingers of an operator, in order to then exert with the reaching-through fingers a merely qualitatively indicated tensile force Z on the actuating component **16** and to remove the actuating component **16** out of its operating position depicted in FIG. **1**.

[0065] The actuating component **16** abuts on its from the hook mouth **24** remotely lying side against the latching formation **20**, which is also a first hook body-side abutment formation **58** in terms of the descriptive introduction. A concavely curved, preferably negative part-cylindrical, first abutment counter-formation **60** nestles in the abutment situation against the convexly curved, preferably part-cylindrical or cylindrical, abutment surface of the first hook body-side abutment formation **58** or of the latching formation **20**, respectively.

[0066] On the side of the actuating component **16** facing towards the hook mouth **24**, the former abuts with a second abutment counter-formation **62** configured in the region of its annular eyelet against a second hook body-side abutment formation **64** configured at the hook body section **38**. Through these abutment engagements and through the pre-tensioning force of the pre-tensioning spring **18**, the actuating component **16** is adequately defined in its operating position and does not move without an external influence.

[0067] In order now to move the quick-coupling hook **10** starting from the operational state shown in FIG. **1** into the setting-up state shown in FIG. **4**, in which the securing lock **14** is in the releasing position and the actuating component **16** is in the setting-up position, an operator after gripping through the annular eyelet **52** pulls at the latter in the direction of the arrow of the tensile force Z and begins to pull the actuating component out of the recess **42** of the hook body **12**. If the annular eyelet **52**, as in the present case, is larger than the diameter of a finger gripping through it, it makes sense to assume that the operator will grip at the section **52***a* of the annular eyelet **52** which essentially runs orthogonally to the desired pulling direction, that is, in the present case orthogonally to the arrow of the tensile force Z.

[0068] The locations of the two spring bearings **22** and **46** of the pre-tensioning spring **18** are so chosen that the virtual straight connecting line **48** which determines the course of the force effect for one thing runs as close as possible past the second hook body-side abutment formation **64** and for another does not deviate too much from the desired direction of the tensile force Z.

[0069] The respective direction of the tensile force Z depends on the respectively gripping operator and cannot be readily visualized from the device. On ergonomic ground, an advantageous direction of the tensile force Z in the reference state does not differ essentially from the orientation direction of the mouth center plane **33** or of the insertion trajectory **32**, respectively, such that at least in the side view of FIG. **1** the virtual straight connecting line **48** encloses an angle with the mouth center plane **33** or the insertion trajectory **32**, respectively, which preferably is not greater than 15°.

[0070] On the side of the hook body **12** facing away from the hook mouth **24**, the former exhibits a connection surface **66** which ordinarily at least in the height direction of the quick-coupling hook **10** is not curved. The connection surface **66** is preferably a connection plane **66**. With this connection surface **66**, the quick-coupling hook **10** for attachment to a vehicle-side carrier is bonded with the latter, for instance butt welded onto the longitudinal end of the latter.

[0071] For the aforementioned reasons, in the reference state the virtual straight connecting line **48** preferably encloses with the connection surface **66** an angle not exceeding 10°.

[0072] Everything said here regarding the virtual straight connecting line **48** also applies to the plane **50** defined above as a reference plane **50** which contains the virtual straight connecting line **48**.

[0073] As becomes clear from the depiction of FIG. **1**, in the reference state the distance D of the virtual straight connecting line **48** from the first hook body-side abutment formation **58** is significantly greater than the distance d of the virtual straight connecting line **48** from the second hook body-side abutment formation **64**. In the depicted embodiment example, the distance D is more than twelve times greater than the distance d. Through this arrangement, a large part of the volume and/or of the mass respectively of the actuating component **16**, in the depicted embodiment example approximately 90% of the volume and of the mass of the actuating component **16**, is situated on a side of the virtual straight connecting line **48** and/or of the plane **50** respectively on which the tensile force Z also engages through the operator. This is also the side on which the latching formation **20**, which in the setting-up state of FIG. **4** secures the actuating component **16** in its setting-up position, is situated.

[0074] The virtual straight connecting line **48** and/or the plane **50** respectively runs outside the reach-through area **56** of the annular eyelet **52**, which further increases the tilt moment about a tilt axis orthogonal to the drawing plane of FIG. **1** effected through the tensile force Z and the pre-tensioning force V.

[0075] Through this design, the tensile force Z and the pre-tensioning force V effect a tilt moment exerted on the actuating component **16**, which pushes the actuating component **16** to tilt anticlockwise in FIG. **1**.

[0076] This tilt moment ensures that even in the case of an operator pulling thoughtlessly and without a defined pulling direction at the annular eyelet **52** with the tensile force Z, after leaving

the operating position a sliding surface **68** at the outer circumference of the actuating component **16** abuts against the latching formation **20** and/or against the first hook body-side abutment formation **58** respectively. The movement of the actuating component **16** when being pulled out of the recess **42** is guided by sliding of the sliding surface **68** along the first hook body-side abutment formation **58** and/or latching formation **20**, respectively.
[0077] The sliding surface **68** which in the depicted embodiment example extends over the thickness of the actuating component **16** runs from the first abutment counter-formation **60** up to an actuating component-side latching counter-formation **70**, which in the setting-up position of the actuating component **16** is in abutment engagement with the latching formation **20**. The actuating component-side latching counter-formation **70** is designed as a negative part-cylindrical shape, such that it can abut flat in a nestling manner against the cylindrical latching formation.
[0078] FIG. **2** shows the quick-coupling hook **10** of FIG. **1** in a transitional phase, after the actuating component **16** has been pulled manually out of the recess **42** to a certain extent out of its operating position shown in FIG. **1**.
[0079] Due to the pullout of the actuating component **16** out of the recess **42**, the pre-tensioning spring **18** is more strongly tensioned than in FIG. **1**, whereby the pre-tensioning force exerted by the pre-tensioning spring **18** on the actuating component **16** has increased in magnitude. In terms of an equilibrium of forces, therefore, the tensile force to be applied by the operator has also become greater. In FIG. **2**, neither tensile force nor pre-tensioning force are recorded but their lines of action are depicted, once in the known form of the virtual straight connecting line **48** and once more through the line of action **72** which in the depicted embodiment example is likewise parallel to the drawing plane of FIG. **1**.
[0080] Due to the tilt moment effected by the tensile force and pre-tensioning force, the actuating component **16** abuts with its sliding surface **68** against the latching formation **20**. On the edge side facing towards the hook mouth **24**, the actuating component **16** still abuts against the second hook body-side abutment formation **64**. The second actuating component-side abutment counter-formation **62** therefore constitutes in the present embodiment example, like the sliding surface **68**, an edge surface usable as a sliding surface running in the thickness direction of the actuating component **16**.
[0081] The hook body-side spring bearing **22** is by its very nature fixed in place with respect to the hook body **12**. A change in position of the two spring bearings **22** and **46** therefore originates solely from the actuating component-side spring bearing **46**. As is discernible in the position of the virtual straight connecting line **48** in FIG. **2**, the virtual straight connecting line **48**, through the pullout of the actuating component **16** from its operating position into the position shown in FIG. **2**, has moved a little away from the second hook body-side abutment formation **64** and moved a little nearer to the latching formation **20**. Nevertheless, the virtual straight connecting line **48** still as before lies nearer to the second hook body-side abutment formation **64**. More than 50% of the volume and/or of the mass respectively, in the present case approximately 80% of the volume and of the mass, of the actuating component **16** still lies on the same side of the virtual straight connecting line **48** on which the latching formation **20** also lies, i.e. on the side of the virtual straight connecting line **48** facing away from the hook mouth **24**. The distance D of the virtual straight connecting line **48** from the latching formation **20** equals now approximately three times the distance d of the virtual straight connecting line **48** from the second hook body-side abutment formation **64**.
[0082] At least 75%, here even 100%, of the reach-through region **56** of the annular eyelet **52** too, still lies on the same side of the virtual straight connecting line **48** as the latching formation **20**.
[0083] Through the pulling out of the actuating component **16**, the securing lock **14** is withdrawn further into the channel **44** and releases a section of the opening region **34** which in FIG. **1** is still occupied.
[0084] In order to move the actuating component **16** into the setting-up position, it has to be pulled

even further out of the recess **42** against the pre-tensioning effect of the pre-tensioning spring **18**. FIG. **3** shows a more pulled-out state.

[0085] FIG. **3** shows a pullout state of the actuating component **16** out of the recess **42** shortly before the actuating component-side latching counter-formation **70** comes into abutment against the hook body-side latching formation **20**. The actuating component **16** is pulled out of the recess **42** so far that now a boundary edge **71** separating the sliding surface **68** from the actuating component-side latching counter-formation **70** is in abutment against the latching formation **20**.

[0086] Through the even stronger pullout of the actuating component **16** out of the recess **42**, the virtual straight connecting line **48** has moved a little further away from the second hook body-side abutment formation **64** towards the latching formation **20**, where however it still lies significantly nearer to the second hook body-side abutment formation **64**. The distance D of the virtual straight connecting line **48** from the latching formation equals in the state shown in FIG. **3** a little more than twice the distance d of the virtual straight connecting line **48** from the second hook body-side abutment formation **64**.

[0087] As before, the pre-tensioning force exerted by the pre-tensioning spring **18** on the actuating component **16** and the tensile force exerted by finger grip at the section **52***a* of the annular eyelet **52** on the actuating component **16**, ensure a tilt moment acting on the actuating component **16** anticlockwise when observing FIG. **3**, which presses the boundary edge **71** between the sliding surface **68** and the latching counter-formation **70** against the latching formation **20**. On the side of the actuating component **16** facing towards the hook mouth **24**, no abutment engagement with the hook body **12** takes place any longer.

[0088] At least 75%, here even at least 80% of the reach-through region **56** of the annular eyelet **52** are still situated on the same side of the virtual straight connecting line **48** as the latching formation **20**.

[0089] A slight further pulling out of the actuating component **16** leads, due to the movement of the boundary edge **71** relative to the latching formation **20** and due to the described tilt moment, to a tilting movement of the actuating component **16** anticlockwise, which is ended only through the abutment of the latching counter-formation **70** against the latching formation **20**. Since the actuating component **16** is connected pivotably with the securing lock **14** about the linkage axis A and since further the securing lock **14** abuts against a surface of the hook body **12** at two places arranged at a distance from one another, the linkage axis A can guide a tilting movement of the actuating component **16** anticlockwise induced by the described tilt moment. This tilting movement would, however, take place in the described manner even if the securing lock **14** were not present. Therefore, regardless of how the operator pulls the actuating component **16** further out of the recess **42**, starting from FIG. **3**, the actuating component **16** will come to abutment with its latching counter-formation **70** against the latching formation **20**.

[0090] The boundary edge **71** does not have to be a sharp edge, but rather can be a boundary region separating the sliding surface **68** from the latching counter-formation **70**.

[0091] The securing lock **14** is almost completely withdrawn into the channel **44**.

[0092] Finally, in FIG. **4** there is depicted the setting-up state of the quick-coupling hook **10** in which the actuating component **16** is in its setting-up position. The latching counter-formation **70** abuts in a nestling manner against the latching formation **20**, and through the pre-tensioning force of the pre-tensioning spring **18** the actuating component **16** is pushed about the latching formation **20** to abutment against the hook body section **38** on its side facing towards the hook mouth **24**. More precisely, a setting-up abutment counter-formation **74** of the actuating component **16** configured on the side of the actuating component **16** facing towards the hook mouth **24** at a distance from the latching counter-formation **70** abuts against a hook body-side setting-up abutment formation **38***b* formed by an area section of the hook body section **38**. The actuating component **16** is again, through the abutment engagements between the latching counter-formation **70** and the latching formation **20** on the one hand and between the setting-up abutment counter-formation **74** and the setting-up abutment formation **38***b* on the other and under the effect of the pre-tensioning spring **18**, in a stable position which can only be changed through an external force application.

[0093] The securing lock **14** is now completely withdrawn into the channel **44** and the opening region **34** is available in its entire clear width for an insertion or a removal of a counter-coupling component into the accommodating region **26** or out of the accommodating region **26**, respectively.

[0094] The virtual straight connecting line **48** has, compared with FIG. **3**, moved even further away from the second hook body-side abutment formation **64** and moved nearer to the latching formation **20**. The virtual straight connecting line **48** preferably never moves, during the entire movement process of a movement of the actuating component **16** out of its operating position into the setting-up position shown in FIG. **4**, nearer to the latching formation **20** than would make the distance d twice greater than the distance D.

[0095] The distance D of the virtual straight connecting line **48** from the latching formation **20** is furthermore approximately as great as or slightly greater than a distance q of the virtual straight connecting line **48** from the hook body-side setting-up abutment formation **38***b*. The distance q, with the given construction of the embodiment example, is slightly greater than the distance d of the virtual straight connecting line **48** from the second hook body-side abutment formation **64**. The distance q preferably also becomes no greater than double, preferably than 1.25 times, the distance D. Thereby there is maintained, through the pre-tensioning spring **18**, an adequate tilt moment about the latching formation **20** acting on the actuating component **16**.

[0096] In all the states of the quick-coupling hook **10** shown in FIGS. **1** to **4**, the virtual linkage axis A is situated on the same side of the virtual straight connecting line **48** as the hook body-side latching formation **20**.

[0097] Out of the securing position shown in FIG. **1** the actuating component **16** does not get into the setting-up position shown in FIG. **4** through pushing of the securing lock **14** by means of a counter-coupling component **30** which is moved into the hook mouth **24**, but only through manual or actuator actuation of the actuating component **16**. If, in contrast, the securing lock **14** is pushed out of its securing position by means of a counter-coupling component **30** being moved into the hook mouth **24**, this pushing out is only temporary. As soon as the counter-coupling component **30** entering the hook mouth **24** has passed the pushed out securing lock **14**, the latter is moved by the pre-tensioning spring **18** back into its securing position.

[0098] By means of a movement of the force application formation **54** in FIG. **4** in the direction away from the hook mouth **24** or more precisely from the mouth center plane **33** and towards the connection surface **66**, the quick-coupling hook **10** can be moved back into the operational or reference state, as the case may be, of FIG. **1**. Due to the physical movement restriction of the securing lock **14** by the hook body **12** shown in FIG. **4**, under the mentioned force load of the actuating component **16** the linkage axis A remains essentially in the position shown in FIG. **4**. The actuating component **16** pivots clockwise—when observing FIG. **4**—about the linkage axis A, whereby the latching counter-formation **70** is no longer in engagement with the latching formation **20**. The pre-tensioning spring **18** then forces the actuating component into its operating position and consequently the securing lock **14** into its securing position.

[0099] While considerable emphasis has been placed on the preferred embodiments of the invention illustrated and described herein, it will be appreciated that other embodiments, and equivalences thereof, can be made and that many changes can be made in the preferred embodiments without departing from the principles of the invention. Furthermore, the embodiments described above can be combined to form yet other embodiments of the invention of this application. Accordingly, it is to be distinctly understood that the foregoing descriptive matter is to be interpreted merely as illustrative of the invention and not as a limitation.

Claims

1-15. (canceled)
16. A quick-coupling hook for agricultural vehicles, comprising i. a hook body with a hook mouth, where the hook mouth exhibits an accommodating region accessible through an opening region, ii. a securing lock which is accommodated at the hook body and which is displaceable relative to the hook body between a securing position and a releasing position, where in the securing position the securing lock projects further into the opening region than in the releasing position, iii. an actuating component which is accommodated at the hook body and which is moveable relative to the hook body between an operating position and a setting-up position, where the securing lock and the actuating component are coupled with one another for common movement in such a way that when the actuating component is in its setting-up position, the securing lock is in its releasing position, and that when the actuating component is in its operating position, the securing lock is in its securing position, iv. a pre-tensioning spring which extends between a hook body-side spring bearing and an actuating component-side spring bearing and which pre-tensions the actuating component towards its operating position, v. a first hook body-side abutment formation on a first side of the actuating component, and vi. a second hook body-side abutment formation on a second side of the actuating component which is opposite to the first side, where the actuating component abuts in its operating position against the first and against the second hook body-side abutment formation, where the actuating component exhibits a force application formation which is configured, through force application thereon, to move the actuating component starting from the operating position against the effect of the pre-tensioning spring in the direction of the setting-up position, where in a reference state of the quick-coupling hook in which the actuating component is in its operating position, a virtual straight connecting line passing through both the hook body-side spring bearing and the actuating component-side spring bearing exhibits at least double the distance from the first hook body-side abutment formation as it does from the second hook body-side abutment formation.
17. The quick-coupling hook according to claim 16, wherein in the reference state of the quick-coupling hook the virtual straight connecting line runs between the first and the second hook body-side abutment formation.
18. The quick-coupling hook according to claim 17, wherein the quick-coupling hook exhibits a hook body-side latching formation with which an actuating component-side latching counter-formation of the actuating component, in a setting-up state of the quick-coupling hook in which the actuating component is in the setting-up position, is engaged in order to hold the actuating component against the effect of the pre-tensioning spring in the setting-up position.
19. The quick-coupling hook according to claim 16, wherein the quick-coupling hook exhibits a hook body-side latching formation with which an actuating component-side latching counter-formation of the actuating component, in a setting-up state of the quick-coupling hook in which the actuating component is in the setting-up position, is engaged in order to hold the actuating component against the effect of the pre-tensioning spring in the setting-up position.
20. The quick-coupling hook according to claim 19, wherein in the reference state of the quick-coupling hook the hook body-side latching formation lies on the same side of the actuating component as the first hook body-side abutment formation.
21. The quick-coupling hook according to claim 20, wherein the hook body-side latching formation is the first hook body-side abutment formation.
22. The quick-coupling hook according to claim 19, wherein the hook body-side latching formation is the first hook body-side abutment formation.
23. The quick-coupling hook according to claim 19, wherein in the setting-up state of the quick-coupling hook a setting-up abutment counter-formation of the actuating component situated at a

distance from the actuating component-side latching counter-formation abuts against a hook body-side setting-up abutment formation, where in the setting-up state the distance of the virtual straight connecting line from the hook body-side setting-up abutment formation equals at most double the distance of the virtual straight connecting line from the latching formation.

24. The quick-coupling hook according to claim 21, wherein in the setting-up state of the quick-coupling hook a setting-up abutment counter-formation of the actuating component situated at a distance from the actuating component-side latching counter-formation abuts against a hook body-side setting-up abutment formation, where in the setting-up state the distance of the virtual straight connecting line from the hook body-side setting-up abutment formation equals at most double the distance of the virtual straight connecting line from the latching formation.

25. The quick-coupling hook according to claim 16, wherein in the reference state the first hook body-side abutment formation is situated on the side of the actuating component facing away from the hook mouth and the second hook body-side abutment formation is situated on the side of the actuating component facing towards the hook mouth.

26. The quick-coupling hook according to claim 16, wherein in the reference state and/or in the setting-up state the virtual straight connecting line encloses an angle not exceeding 15° a) with a mouth center plane which is oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth and running centrally both through the opening region and through the accommodating region, and/or b) with a virtual insertion trajectory of a counter-coupling component to be coupled with the quick-coupling hook into the accommodating region.

27. The quick-coupling hook according to claim 16, wherein the hook body exhibits a connection surface for connecting to a vehicle-side carrier, where in the reference state and/or in the setting-up state the virtual straight connecting line encloses with the connection surface an angle not exceeding 10°.

28. The quick-coupling hook according to claim 19, wherein the actuating component exhibits a sliding surface which runs from a first abutment counter-formation with which in the reference state the actuating component abuts against the first hook body-side abutment formation up to the actuating component-side latching counter-formation.

29. The quick-coupling hook according to claim 16, wherein the securing lock is articulated pivotably about a virtual linkage axis at the actuating component.

30. The quick-coupling hook according to claim 19 wherein the securing lock is articulated pivotably about a virtual linkage axis at the actuating component, wherein in the setting-up state the virtual linkage axis lies on the same side as the hook body-side latching formation and/or the first hook body-side abutment formation with respect to a reference plane containing the virtual straight connecting line and oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth.

31. The quick-coupling hook according to claim 16, wherein the force application formation comprises a rear grip formation for a manual finger grip and/or that the force application formation comprises a coupling formation of the actuating component with an output element of an actuator.

32. The quick-coupling hook according to claim 19, wherein in the reference state at least 75% of the actuating component lie on the same side with respect to a reference plane containing the virtual straight connecting line and oriented orthogonally to a cross-sectional area of the accommodating region surrounded by the hook mouth as the hook body-side latching formation and/or the first hook body-side abutment formation.

33. The quick-coupling hook according to claim 16, wherein the quick-coupling hook exhibits an actuator in order to move the actuating component from the operating position into the setting-up position.